ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

LAB09: PULP-NE16

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Objective of the Class

Lesson: NE16 accelerator

Programming Language: C

Lab duration: 3h

Assignment:

- Time for delivery: 2 weeks

Deadline:

Dec 26th 2025

<u>The class is meant to be interactive: coding together, on your own, and do not be afraid to ask questions!</u>

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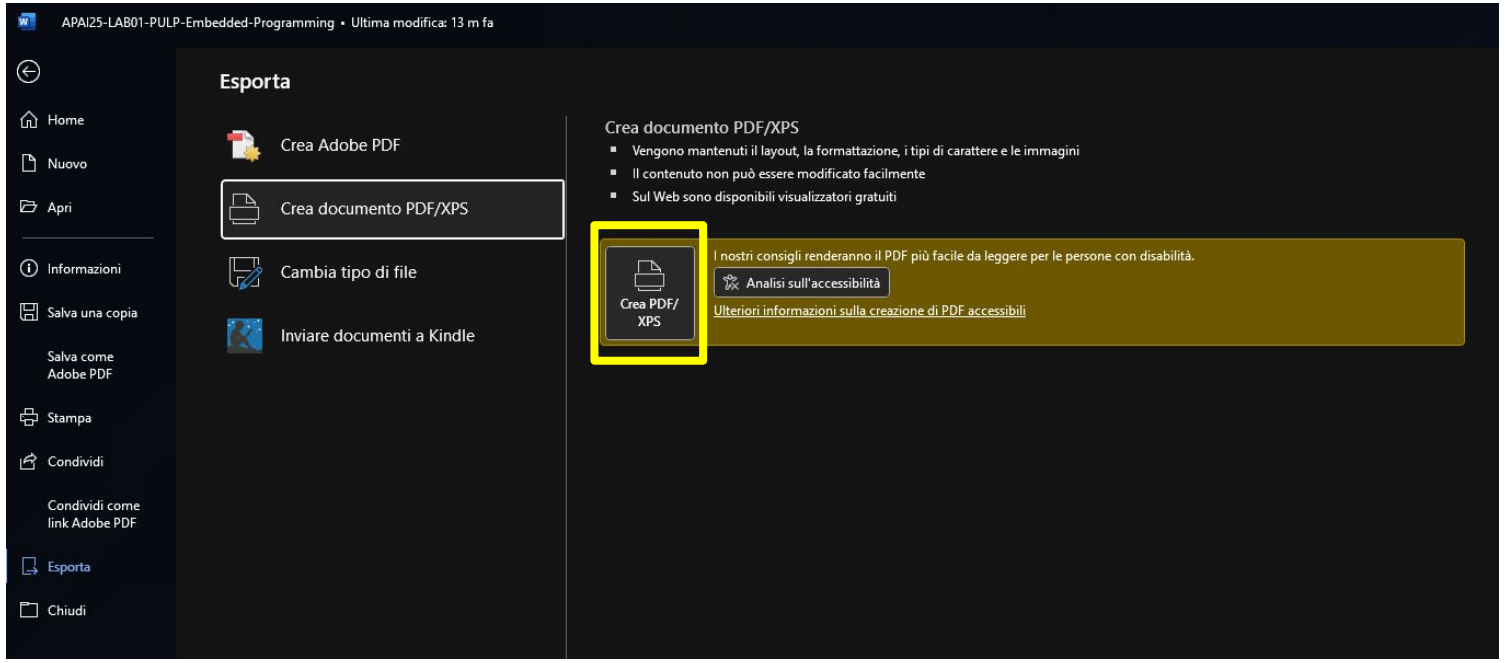
How to deliver the Assignment

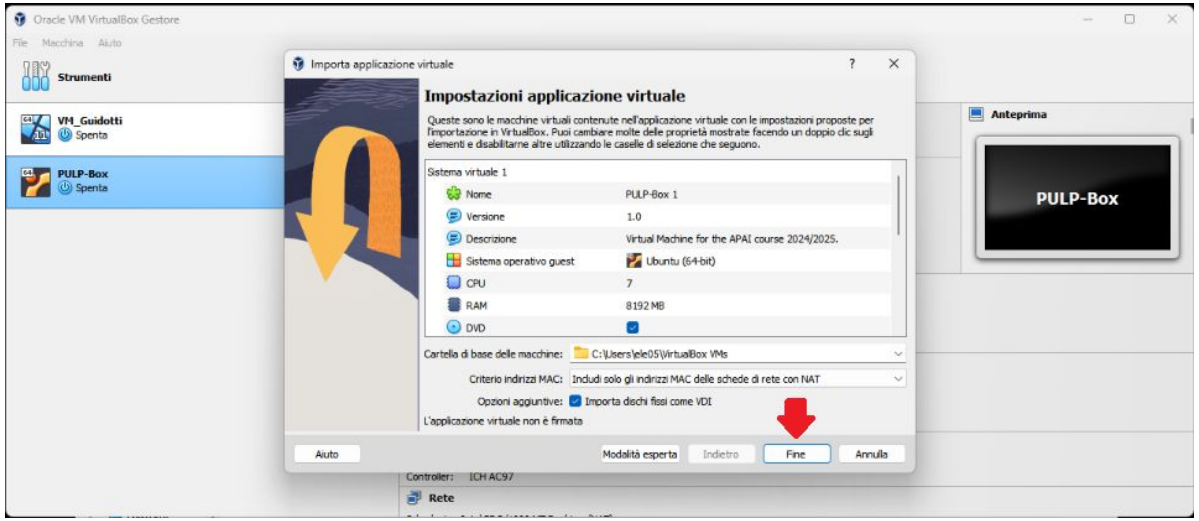
You will deliver ONLY the PDF assignment, no code

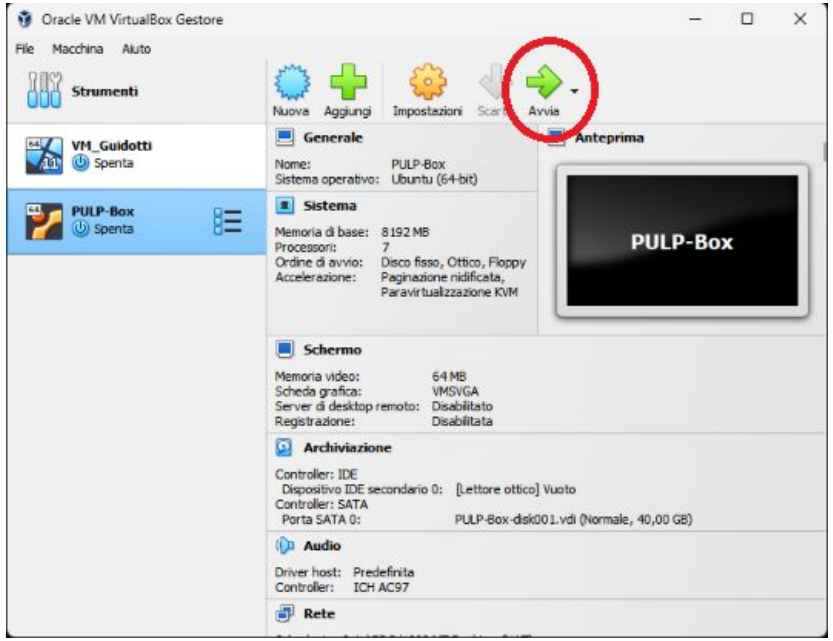
- Download the assignment file from Virtuale.
- Fill the results required by the assignment.
- Export to pdf format.
- Rename the file to: LAB<number_of_the_lesson>_APAI_<your_name>.pdf
- Use Virtuale platform to load ONLY your .pdf file



Opening the VM and VSCode

1. On the lab's PCs, open the file explorer and go to This PC, C:/VM_APAI/
2. Double click on PULP-box.ova
3. VirtualBox opens, just click on "Fine"
4. Wait for the VM to be imported
5. Open the VM with "Avvia"





Password is 'pulp'

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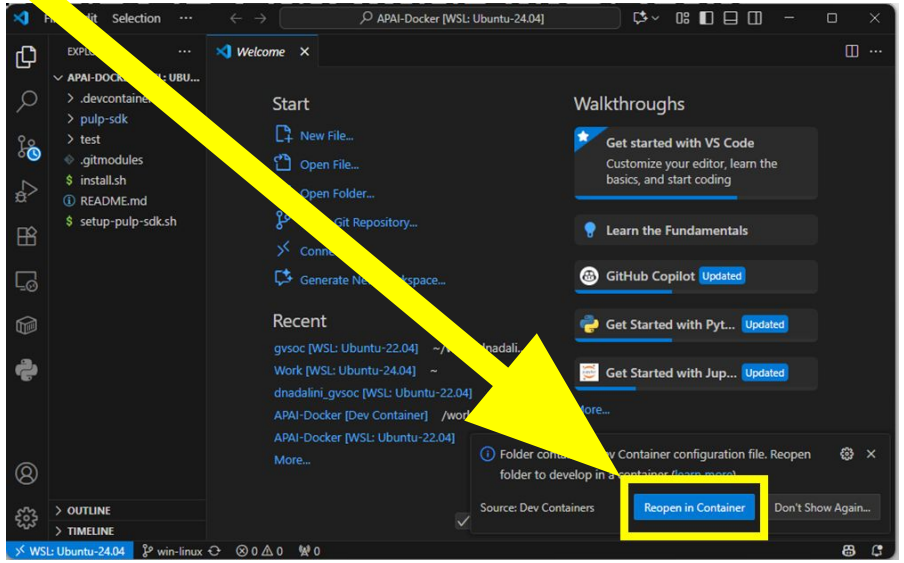
Opening the Docker with VSCode

1. Open a terminal (right click – open a new terminal)
2. From the terminal, open VSCode in the folder of the Docker

```
$ cd APAI-Docker
$ code .
```

3. Reopen the APAI-Docker folder in VSCode (click on “Reopen in container”)
4. Now you can use the integrated terminal (open with CTRL+J) to run your applications!



IMPORTANT: every time you open a new terminal to work on PULP, launch

```
$ source setup-pulp-sdk.sh
```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Getting Started

IMPORTANT: activate the pulp-sdk module file <u>every</u> time a new shell is open.

```
$ source setup-pulp-sdk.sh
```

For the lab of today, we need to:

```
$ pip install mako numpy six
```

HOW TO RUN THE CODE:

```
$ git clone https://github.com/EEESlab/APAI25-LAB09-NE16
$ cd APAI25-LAB09-NE16
$ python3 parameters_generate.py --kernel-shape=<add_here>
--channels-in=<add_here> --channels-out=<add_here>
--output-spatial-dimension=<add_here>
$ make clean all run
```


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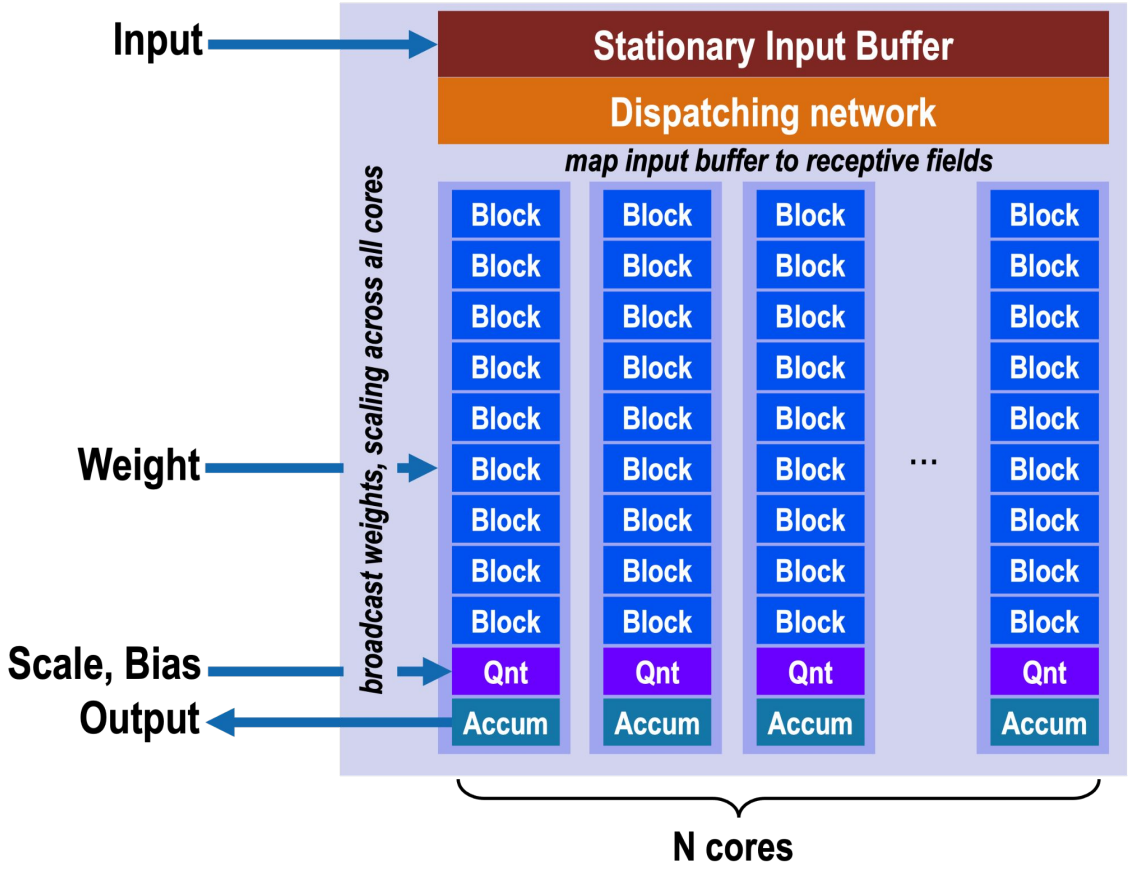
Theory recap

Here are the lecture slides on NE16



Prerequisites/how to run code

Prerequisites:

```
source setup_pulp_sdk.sh
```

Run the code:

```
1.  python parameters_generate.py --kernel-shape=<add_here> --channels-in=<add_here> --channels-out=<add_here>
    --output-spatial-dimensions=<add_here>
2.  make clean all run
```

Follow the assignment document.

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